



3D convolutional neural network-based one-stage model for real-time action detection in video of construction equipment

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Abstract

This study aims to propose a three-dimensional convolutional neural network (3D CNN)-based one-stage model for real-time action detection in video of construction equipment (ADVCE). The 3D CNN-based single-stream feature extraction network and detection network are designed with the implementation of the 3D attention module and feature pyramid network developed in this study to improve performance. For model evaluation, 130 videos were collected from YouTube including videos of four types of construction equipment at various construction sites. Trained on 520 clips and tested on 260 clips, ADVCE achieved precision and recall of 82.1% and 83.1%, respectively, with an inference speed of 36.6 frames per second. The evaluation results indicate that the proposed method can implement the 3D CNN-based one-stage model for real-time action detection of construction equipment in videos of diverse, variable, and complex construction sites. The proposed method paved the way to improving safety, productivity, and environmental management of construction projects.

1 | INTRODUCTION

Recently, as the “global lockdown” is demanding minimal human contact and digital communications amid the coronavirus disease 2019 (COVID-19) pandemic (Lewnard & Lo, 2020; Spinelli & Pellino, 2020), the fourth industrial revolution that had started in all sectors is further accelerating (AlMaadeed, 2020; Costabile et al., 2020). Despite the need for reduced human contact, the labor-intensive construction industry shows a relatively lower teleworking rate (22%) than other industries (Dingel & Neiman, 2020) and still regards face-to-face contact as an essential requirement. Therefore, limitations manifest in remote monitoring and control of worksites (Y. Fang et al., 2016; Golparvar-Fard et al., 2013). Accordingly, the importance of automated and unmanned operation

in construction projects has become more obvious than ever (Björck et al., 2020). In the future, smart construction that uses emerging technologies (e.g., artificial intelligence, big data, etc.) will become essential in construction sites to strengthen remote monitoring and management and to promote off-site construction in the future (Maskuriy et al., 2019). These include (i) work safety—by preventing safety accidents through the recognition of the movement of construction equipment in real-time (W. Fang et al., 2020); (ii) construction productivity—by monitoring equipment in real-time so as to increase marginal product labor (Goodrum et al., 2011; Zhai et al., 2009); and (iii) environmental management—environmental management of pollutants (noise, vibration dust, and greenhouse gases) produced by construction equipment operation (Jung et al., 2019, 2020; Kang et al., 2020).



Construction equipment monitoring is traditionally performed directly by a construction manager and takes much time and effort to simultaneously cover all sites in a large construction project (J. Kim & Chi, 2019). Accordingly, previous studies have developed methods to monitor the movement and location of construction equipment using sensors. In some studies, sensor-based monitoring methods for construction equipment using various sensors such as radio frequency identification systems (RFID; Y. Fang et al., 2016), global positioning systems (GPS; Alshibani & Moselhi, 2016), light detection and ranging (Wang et al., 2015), mobile phone (Akhavian & Behzadan, 2015), and inertial measurement units (H. Kim, Ahn, et al., 2018) have been developed. However, such methods require sensors for individual construction equipment. Furthermore, location-based construction equipment monitoring methods based on GPS or RFID are limited in that they do not recognize the type of construction equipment and action.

On the other hand, other studies monitored construction equipment in real-time by using computer vision-based technology that collected and analyzed construction site information automatically (Golparvar-Fard et al., 2013; Heydarian et al., 2012). Since the computer vision-based method does not require additional sensors, except for the device for image capture, it is both time- and cost-efficient (Roberts & Golparvar-Fard, 2019). Recently, several studies that used the convolutional neural network (CNN) based on big data have been conducted in two categories (W. Fang et al., 2018; Roberts & Golparvar-Fard, 2019; Yan et al., 2020): (i) Object detection, which combined object localization and classification; and (ii) action detection, which can recognize and classify the motion of equipment or workers. The object detection model predicts the type of construction equipment and its location using the image-based two-dimensional (2D) CNN (e.g., Faster region-based CNN (R-CNN) and Single shot multibox detector (SSD); W. Fang et al., 2018; Thakar et al., 2018). However, 2D CNN only extracts spatial features and is inappropriate for classification or detection of the actions that require temporal features from a sequence of images. Other studies proposed multistage action detection methods where the action classification stage was performed using the long short-term memory (LSTM) or three-dimensional (3D) CNN to extract the spatio-temporal features, which represent the time-series change of the spatial features extracted from the object detection stage (Chen et al., 2020; Luo et al., 2020). However, each stage of the multistage action detection method produces errors and requires considerable processing time. Thus, such errors at the object detection stage may cause a huge impact on accuracy in the action classification stage, consequently making real-time detection difficult.

This study aims to propose ADVICE (action detection in video of construction equipment) for monitoring actions of construction equipment at various construction sites. ADVICE is a 3D CNN-based one-stage model for real-time action detection of construction equipment. Toward this end, the 3D CNN-based single-stream feature extraction network that extracts the spatio-temporal features and the detection network that generates the detection results were designed. Also, the 3D attention module, feature pyramid network, and multiclass output algorithms were developed to improve the model performance while maintaining the inference speed for real-time detection. For model application at complex and various construction sites, the performance of the ADVICE was analyzed, and its reliability was evaluated using the video dataset of the multiple construction sites collected from YouTube.

2 | RELATED WORKS

In this section, various related works on the monitoring of construction sites using a computer vision-based method were reviewed. Focusing on the object and action detection, this study reviewed previous studies on the 2D CNN that is used based on the emerging big data and computing-related technologies and the 3D CNN, which is being used effectively in video processing recently.

2.1 | 2D CNN

With the recent advancements in computer calculation performance, the 2D CNN has shown much improved performance in processing big data compared to the traditional hand-crafted feature method. As a result, the 2D CNN is now widely used in studies related to image and video processing tasks. Also in the construction industry, several studies have developed a model that uses the 2D CNN to detect objects in a construction site for structural health monitoring (Cha et al., 2018; R. Li, Yuan, et al., 2018; Liang, 2019; Narazaki et al., 2018) or workers' safety (W. Fang et al., 2018; Yan et al., 2020).

Several studies have been conducted not only on structural health monitoring and workers' safety but also on construction equipment monitoring. Roberts and Golparvar-Fard (2019) proposed an action labeling method using the hidden Markov model and proposed end-to-end detection and tracking method based on the 2D CNN. Object detection was performed in real-time by calculating the frame within 1 s with the Faster R-CNN (W. Fang et al., 2018). H. Kim, Kim, et al. (2018) conducted object detection of construction equipment using the region-based fully convolutional network (R-FCN). Yan et al. (2020)



proposed the method to detect the 3D information of an object (top, front, and elevation) by inputting 2D images from the camera to the R-FCN. Arabi et al. (2020) conducted the research on the development of solutions for construction vehicles by proposing an object detection model that used an improved version of SSD MobileNet.

In several studies focusing on action detection, which requires additional temporal features, 2D CNN is also used to recognize and classify actions. Luo et al. (2018) classified the action of workers by tracking their work using the multi-domain CNN and by extracting the spatial and temporal features based on the optical flow estimation technology. Slaton et al. (2020) proposed a classification framework for the action of construction equipment by introducing a hybrid network between the LSTM and the 2D CNN. Kim and Chi (2020) conducted an interaction analysis between actions after classifying their visual features and operation cycles with the 2D CNN and the double-layer LSTM.

At the construction site, 2D CNN, such as Faster R-CNN, conducted fast object detection of construction equipment, which is spatial information, and maintained high accuracy in action recognition. However, 2D CNN can only deal with 2D information, which is related to the spatial features. Therefore, for action detection using the 2D CNN, other algorithms for encoding spatio-temporal information, such as LSTM, are required. This entails additional computation so action detection cannot be performed in real time.

2.2 | 3D CNN

For action detection, spatio-temporal features, which are the combination of the spatial features required for object detection and the temporal features required for action classification, should be extracted. The 3D CNN uses the 3D convolutional layer and the 3D pooling layer that adds the depth axis to the 2D CNN. As sequential video frames are stacked along with the depth axis, the spatio-temporal features can be extracted. The 3D CNN showed considerably higher performance than the 2D CNN in terms of action classification and detection tasks (Ji et al., 2013). In particular, the 3D CNN is widely used in the classification or detection of human actions (R. Zhao et al., 2017) and hand gestures on videos (Ge et al., 2017).

In keeping with such a trend, the construction industry also used the 3D CNN to develop an action classification or action detection method in order to monitor construction workers and equipment action. Chen et al. (2020) analyzed the productivity of construction equipment by using the 3D CNN after implementing the Fast R-CNN. Luo et al. (2020) improved the action classification accuracy by com-

binning deep activity features and contextual information after having detected objects with the YOLOv3 and classified their action with the 3D CNN. These two studies proposed a multistage framework consisting of the object detection, tracking, and action detection stages. First, the object detection was performed with extracted spatial features from the 2D CNN. Through a tracking algorithm, the sequential spatio-temporal features were extracted as to be input to the 3D CNN. Finally, the action classification result from the 3D CNN was combined with the object detection result.

The proposed multistage framework could not perform the whole process from the spatio-temporal feature extraction to the action detection at once. Even though the studies processed the action classification individually at a speed of 15 frames per second (fps) or higher, the whole process of the object detection and the action classification could not be performed in real time. Furthermore, since the errors generated in the previous stage would affect the following stage in a multistage framework, the performance of action classification will be degraded due to the errors in object detection and tracking.

3 | RESEARCH SCOPE AND CHALLENGES

In the field of computer science, there has been active research on the 3D CNN-based one-stage model, which can extract spatial and temporal features at the same time and achieved high performance in human action detection (Carrara et al., 2019; Hou et al., 2017; Kalogeiton et al., 2017; Z. Li, Gavriluk, et al., 2018; J. Zhao & Snoek, 2019). Compared to the multistage model, the one-stage model offers fast computation and requires smaller datasets as its object detection and action classification is performed end-to-end. In particular, the single-stream model, where input data for action classification and object detection are entered together, has a simple architecture and requires less computation.

Despite such advantages of the 3D CNN-based one-stage action detection model, no studies used it for construction equipment action detection in real time. Therefore, it is necessary to apply the 3D CNN-based one-stage action detection model to the real-time monitoring of construction equipment actions. Before it can be applied, however, the following challenges must be resolved.

First, there are diverse types of construction equipment and corresponding equipment actions. The existing 3D CNN-based one-stage model targets human actions and predicts only the action class. However, this study needs to predict the object class, that is, the type of construction equipment and the action class of the

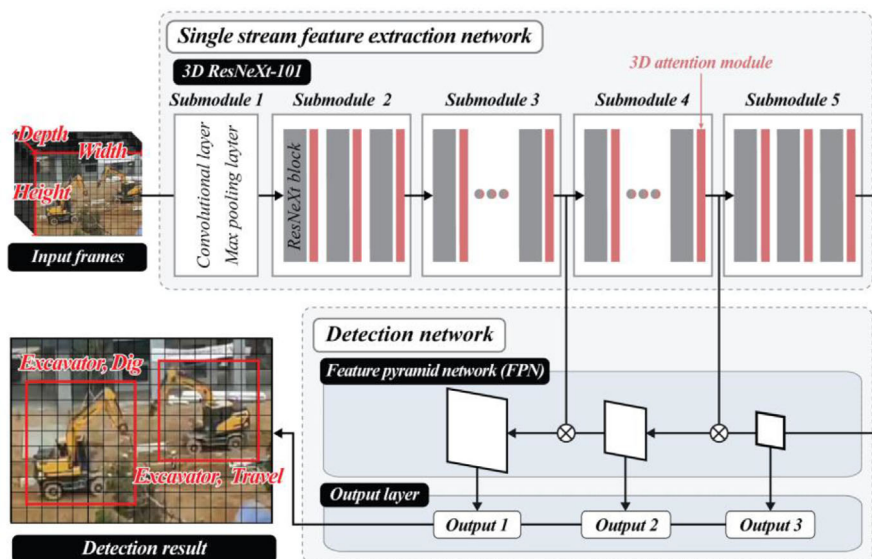


FIGURE 1 Overview of the developed model, action detection in video of construction equipment (ADVICE)

corresponding construction equipment. Second, at a construction site, besides construction equipment, there are various other objects, such as materials and structures in the background, and each construction site has different backgrounds. Furthermore, construction equipment and their actions in the same class have differences such as colors, sizes, and moving speed, whereas among classes, differences are insignificant such as idling and traveling, which can be performed by all the construction equipment. Therefore, there is a high similarity among classes. Thus, it is important to segment the construction equipment from its background and extract significant spatio-temporal features from similar appearances (e.g., having wheels and a bucket) and actions (e.g., digging and unloading soil). Finally, multiple number of equipment perform various activities in construction sites at the same time, and their scales in video frames are diverse. Since construction equipment are used in diverse locations at a construction site, their size in video frames may differ according to their distance from the camera and the optical zoom of the camera, even if they have the same size. Therefore, the model should be able to detect at once multiple numbers of construction equipment that have different scales.

To address the above challenges, ADVICE was developed for real-time action detection of construction equipment (i.e., localization and classification of various types of equipment actions) on diverse videos recorded at different construction sites from various perspectives and distances with varying camera motions. For this whole process to be performed in real time, the inference speed of the model should comply with the recording speed of the video (25 fps or above).

4 | METHODOLOGY

As shown in Figure 1, ADVICE consists of two parts: (i) The single-stream feature extraction network is for extracting spatio-temporal features at each of the three submodules (submodules 3, 4, and 5) from a sequence of video frames after they go through a network in which 3D attention modules are applied to the existing 3D CNN architecture. (ii) The detection network extracts three detection results from the spatio-temporal features, which are taken from the single-stream feature extraction network, after the feature pyramid network and output layers.

4.1 | Single-stream feature extraction network

This study designed a 3D CNN-based single-stream feature extraction network that extracts spatio-temporal features from the input video frames, which is necessary for the detection of construction equipment actions. In extracting spatio-temporal features, the 3D convolutional layer needs input as a four-dimensional (4D) feature map, that is, $F_{\text{input}} \in \mathbb{R}^{W \times H \times D \times C}$ (W , H , D , and C are number of width, height, depth, and channels, respectively), which generates an output feature map (F_{output}) after 3D convolution computation (Diba et al., 2018). Particularly, the 3D attention module was developed to improve action detection performance with slightly increasing computation by focusing on the significant features along both the spatial and temporal dimensions.

TABLE 1 Architecture of the 3D ResNeXt-101 (input: $224 \times 224 \times 16 \times 3$)

Submodule	3D ResNeXt-101 layers	Output feature maps
Submodule 1	$7 \times 7 \times 7, 64, 2 \times 2$ spatial stride, 1 temporal stride	$56 \times 56 \times 8 \times 64$
	$3 \times 3 \times 3$ max pooling, stride 2	
Submodule 2	$\begin{pmatrix} 1 \times 1 \times 1 \ 64 \\ 3 \times 3 \times 3 \ 64 \\ 1 \times 1 \times 1 \ 256 \end{pmatrix} \times 3$	$56 \times 56 \times 8 \times 256$
Submodule 3	$\begin{pmatrix} 1 \times 1 \times 1 \ 128 \\ 3 \times 3 \times 3 \ 128 \\ 1 \times 1 \times 1 \ 512 \end{pmatrix} \times 4$	$28 \times 28 \times 4 \times 512$
Submodule 4	$\begin{pmatrix} 1 \times 1 \times 1 \ 256 \\ 3 \times 3 \times 3 \ 256 \\ 1 \times 1 \times 1 \ 1,024 \end{pmatrix} \times 23$	$14 \times 14 \times 2 \times 1024$
Submodule 5	$\begin{pmatrix} 1 \times 1 \times 1 \ 512 \\ 3 \times 3 \times 3 \ 512 \\ 1 \times 1 \times 1 \ 2,048 \end{pmatrix} \times 3$	$7 \times 7 \times 1 \times 2048$

4.1.1 | 3D CNN architecture

The performance of the feature extraction network affects the accuracy of the overall model (Kornblith et al., 2019). Therefore, 3D ResNeXt-101, one of the single-stream 3D CNN architectures that showed excellent performance in the existing action classification and detection, was used as the basic architecture of the feature extraction network (Hara et al., 2018).

The 3D ResNeXt-101 in this study consists of five submodules (see Table 1). The input size was defined by multiplying 224×224 by the depth (i.e., the number of input frames), following previous studies that used the same size (Köpüklü et al., 2019; Luo et al., 2018).

4.1.2 | 3D attention module

To more accurately detect the construction equipment actions with high-class similarity at a construction site with complex backgrounds, it is necessary to focus on the significant features, which are related to each object and action. The attention mechanism, which has been proven to effectively improve accuracy in various computer vision tasks, can refine input feature maps and emphasize the significant features. Particularly, in the case of an action associated with surrounding objects, such as digging and unloading soil, the attention mechanism can help distinguish its class considering the entire context of the video frames. Several 3D attention modules that refine spatio-temporal features were developed in previous studies; however, those modules considerably increased computational load and time (J. Li et al., 2020; Zhou et al., 2019). Therefore, inspired by the convolutional block attention module (CBAM), which consisted of 2D attention blocks

for spatial feature attention with a negligible overhead of computation (Woo et al., 2018), this study developed the 3D attention module to help refine spatio-temporal features for detection of construction equipment actions by utilizing minimal computational load.

The developed 3D attention module can refine spatio-temporal features by expanding the dimension of the processed 4D features (i.e., the width, height, depth, and channels). Particularly, the module is modularized into simple pooling and convolution so that it could be applied without transforming the existing 3D CNN architecture, such as the 3D ResNeXt-101. The attention module consists of two types of the attention submodule (see Figure 2): (i) The one-dimensional (1D) channel attention module that can emphasize object features; and (ii) the 3D spatio-temporal attention module that can emphasize spatio-temporal features.

As for the 1D channel attention module, it performs the average pooling and max pooling ($AvgPool_{\text{channel}}$, $MaxPool_{\text{channel}}$) on the feature map ($F \in \mathbb{R}^{W \times H \times D \times C}$) along the channel direction (i.e., depth-axis) and passes the result to the multilayer perceptron and the sigmoid function (σ ; see Equation 1). Through the element-wise multiplication of the extracted $M_{\text{channel}}(F) \in \mathbb{R}^{W \times H \times D \times 1}$ and F , the channel refined feature map ($F' \in \mathbb{R}^{W \times H \times D \times C}$) is generated (see Equation 2). As a result, the channel attention module can emphasize the meaningful information as each channel acts as a feature extractor of the given input video frames. As for the 3D spatio-temporal attention module, it performs the average pooling ($AvgPool_{\text{spatio-temporal}}$) and max pooling ($MaxPool_{\text{spatio-temporal}}$) on F' along the spatio-temporal direction (width, height, and depth) and passes the result to the convolution ($Conv$) with filter size $7 \times 7 \times \text{depth}$ and σ (see Equation 3). Through the element-wise multiplication

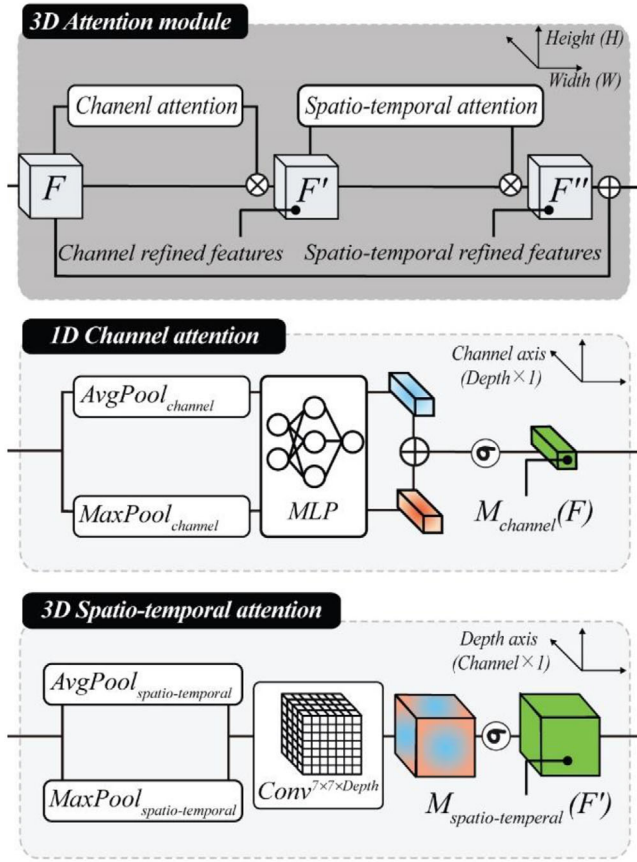


FIGURE 2 3D attention module developed in this study

tion of the extracted $M_{\text{spatio-temporal}}(F') \in \mathbb{R}^{1 \times 1 \times 1 \times C}$ and F' , the spatio-temporal refined feature map ($F'' \in \mathbb{R}^{W \times H \times D \times C}$) is finally generated (see Equation 4). Through this process, 3D spatio-temporal attention can emphasize the context of construction equipment and their surroundings in a construction site since the features were extracted from the width, height, and depth of the given input video frames:

$$M_{\text{channel}}(F) = \sigma(\text{MLP}(\text{AvgPool}_{\text{channel}}(F))) \\ (+ \text{MLP}(\text{MaxPool}_{\text{channel}}(F))) \quad (1)$$

$$F' = F \otimes M_{\text{channel}}(F) \quad (2)$$

$$M_{\text{spatio-temporal}}(F') = \sigma \left(\text{Conv}^{7 \times 7 \times \text{depth}} \left(\begin{bmatrix} \text{AvgPool}_{\text{spatio-temporal}}(F') \\ \text{MaxPool}_{\text{spatio-temporal}}(F') \end{bmatrix} \right) \right) \quad (3)$$

$$F'' = F' \otimes M_{\text{spatio-temporal}}(F') \quad (4)$$

where σ is the sigmoid function. M_{channel} and $M_{\text{spatio-temporal}}$ correspond to the 1D channel attention and the 3D spatio-temporal attention, respectively. $\text{AvgPool}_{\text{channel}}$, $\text{MaxPool}_{\text{channel}}$, $\text{AvgPool}_{\text{spatio-temporal}}$ and $\text{MaxPool}_{\text{spatio-temporal}}$ correspond to the average pooling and the max pooling in the M_{channel} and $M_{\text{spatio-temporal}}$, respectively. F , F' , and F'' refer to the first input feature maps, the channel refined feature maps, and the spatio-temporal-refined feature maps, respectively.

The developed 3D attention module was added at each ResNeXt block in the 3D ResNeXt-101. Similar to the CBAM, it was applied in the order of channel attention module followed by spatio-temporal module.

4.2 | Detection network

A detection network was designed to generate the detection results from the spatio-temporal features. The 2D CNN-based detection network consists of the feature pyramid network and the output layer. The feature pyramid network, which is effective for the detection of multiple construction equipment with different scales, was transformed so that it could be linked to the third, fourth, and last submodules of the feature extraction network to extract spatio-temporal features. In addition, the output layer, which was designed to solve multiclass problems at the end of ADVICE, was designed to generate the results from the features of the three different scales extracted from the feature pyramid network.

4.2.1 | Feature pyramid network

In the existing 2D CNNs, to improve the accuracy of small object detection, the feature pyramid network is used to expand the low-resolution feature map with high-level semantic information (i.e., upsampling) and incorporating low-level semantic information by stage so as to generate a high-resolution feature map with rich semantic information (Lin et al., 2017). Therefore, this study proposes a new feature pyramid network, in which 4D feature maps can be input by adding connection 3D CNN blocks. The new feature pyramid network was applied to the detection network to detect several construction equipment at different scales at the same time (see Figure 3).

In general, feature extraction in the feature pyramid network is performed in the process of bottom-up and top-down pathways. First, the bottom-up pathway computes the high-level semantic feature map as input feature maps

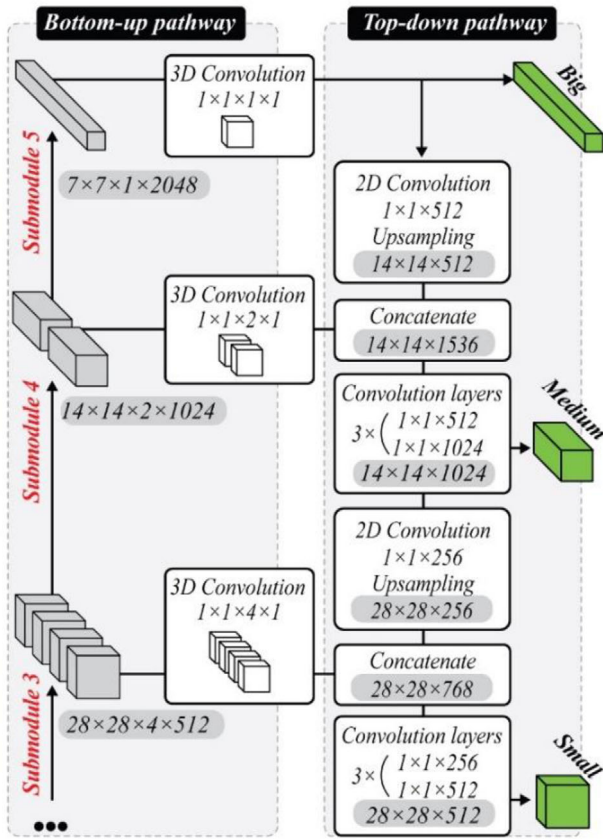


FIGURE 3 Architecture of feature pyramid network; (a) bottom-up pathway (b) top-down pathway

are passed through several submodules. In this study, the bottom-up pathway is designed to generate three different 4D feature maps. Here, the convolutional layer with the filter size of $1 \times 1 \times D \times 1$ was added to compress the depth axis of the 4D feature map. Second, the top-down pathway creates a high-resolution feature map by concatenating the upsampled high-level semantic feature map and the low-level semantic feature map. Toward this end, the feature maps extracted from the third and the fourth submodules of the feature extraction network ($28 \times 28 \times 4 \times 512$ and $14 \times 14 \times 2 \times 1024$) are passed through the convolutional layer with the filter size of $1 \times 1 \times D \times 1$ so as to compress the depth-axis and concatenate it with the upsampled feature map. Finally, the concatenated feature maps are passed through the convolutional layers consisting of three sequences of 1×1 layers to reduce the aliasing effect due to upsampling.

As a result, the feature maps extracted from the three submodules can generate feature maps with three different scales via the feature pyramid network: (i) Feature maps with size $28 \times 28 \times 512$ for small object detection; (ii) $14 \times 14 \times 1024$ for medium object detection; and (iii) $7 \times 7 \times 2048$ for big object detection.

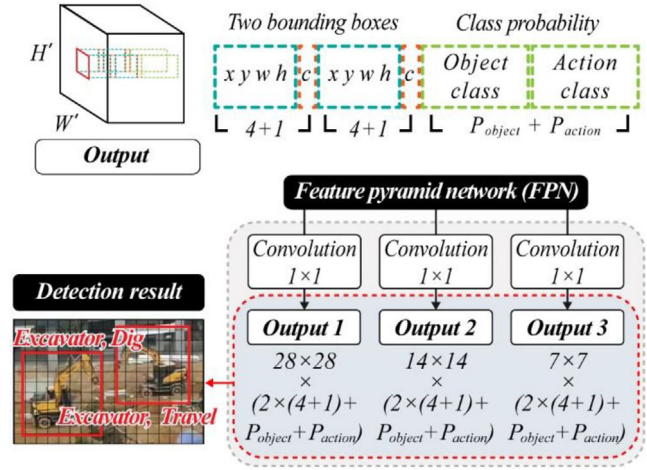


FIGURE 4 Final output of the detection network

4.2.2 | Output layer

The output layers of three different sizes, located at the end of the detection network, consist of 1×1 convolutional layers and output the results based on the bounding box regression guideline of YOLOv3. The output layer predicts the two bounding boxes and the probability of classes of the corresponding grid cell from the video frame divided by $W' \times H'$. In particular, output layers were designed to solve the newly defined multiclass problem (i.e., the object class and the action class) in this study (see Figure 4).

First, the information of each bounding box has five values and consists of four location elements and one confidence score. The location elements are center x -coordinate (x), center y -coordinate (y), width (w), and height (h) of the bounding box, and the confidence score (c) is between 0 and 1, showing the possibility of the object inside the corresponding bounding box.

Second, the probability of classes means the probability that the object in the grid belongs to each class and is between 0 and 1 per class. The existing 3D CNN-based one-stage model was developed for the general human action detection problem, which is a binary-class problem on the action class without distinction of the object class. However, the possible types of action are limited depending on the type of construction equipment. Meanwhile, some actions such as “idling” and “traveling” can be performed by different types of construction equipment. Therefore, instead of treating each object and action class combination as a new binary class, in this study, the problem was defined as a multiclass problem to predict the type of construction equipment (i.e., the object class, P_{object}) and the class of its action (i.e., its action class, P_{action}), respectively, and one class in each object class and action class with the highest score could be determined as a final output. Here, the probability of the action classes that could not be



performed by the predicted type of construction equipment with the highest probability among the object classes was set at 0 by post-processing.

Finally, the output resulting from one output layer is in a 3D cube of $W' \times H' \times (2 \times (4 + 1) + P_{\text{object}} + P_{\text{action}})$. The outputs from the three output layers are combined to indicate all bounding boxes and classes in the single video frame.

5 | EXPERIMENTS

5.1 | Dataset

Since a set of sequential frames was given to the developed ADVISE as an input to predict the bounding box and types of construction equipment and actions, video datasets were collected and annotated for training and testing ADVISE. As shown in the UCF 101 (Soomro et al., 2012) or THUMOS (Idrees et al., 2017) where the video dataset on various human action classes is established, this study used the video dataset that included various construction equipment action classes so as to train and test ADVISE.

First, the types (i.e., excavator, truck, loader, and roller) of construction equipment and their actions (i.e., idle, travel, rotate, dig, and unload) to be detected were determined. Several previous studies on the action of the construction equipment only targeted excavators and trucks (H. Kim, Bang, et al., 2018; Roberts & Golparvar-Fard, 2019), but this study deals with the four types of object classes by additionally including loaders and rollers. In addition, action classes were determined by focusing on the main works of each type of construction equipment and similar actions were considered as the same action in order to establish the standard action of construction equipment (Roberts & Golparvar-Fard, 2019). For instance, moving materials or flattening the land, which are secondary works of excavators, were excluded, and similar swinging buckets and rotating of excavators were all considered as “rotating.” Meanwhile, since digging and unloading of the excavator and the loader have different characteristics in their bucket movements, “dig” and “unload” of excavator and loader were separately classified. Therefore, a total of seven action classes were determined.

Second, the videos of construction equipment actions in various construction sites were collected from YouTube. To verify the robustness of the developed ADVISE, the videos that include various challenging factors of computer vision-based data such as occlusion, lighting, the various scale of objects, and motions of the camera were collected (see Figure 5). From the collected video frames with different durations, unnecessary parts that had no relation



FIGURE 5 Examples of data collected from “Youtube” and challenges related to object recognition: (a) occlusion, (b) lighting, (c) scale of object, and (d) motion of camera including various construction equipment

TABLE 2 Statistics of the annotated construction equipment actions

Equipment (object #)	Actions (action #)	No. of bounding boxes
Excavator (1)	Idle (1)	11,720
	Travel (2)	3843
	Rotate (3)	7835
	Dig (4)	8815
	Unload (5)	6437
Truck (2)	Idle (1)	32,141
	Travel (2)	18,254
Loader (3)	Idle (1)	3021
	Travel (2)	13,512
	Dig (6)	1657
	Unload (7)	5676
Roller (4)	Idle (1)	3620
	Travel (2)	12,811
Total		129,342

to the action of construction equipment (e.g., video logo) or sudden cuts in the video were excluded (Caba Heilbron et al., 2015). A total of 130 videos with 640×480 resolution and 25 sampling rates (a total of 62,854 frames) that contain the movement of the four types of equipment at construction sites were collected.

Third, all 130 videos were annotated. The annotation included bounding box drawing, object class labeling, and action class labeling as the results from the output layer. The annotation process used the open-source program that helps draw bounding boxes using object detection and tracking algorithms (<https://github.com/alexandre01/UltimateLabeling>). The bounding boxes were then adjusted manually in more detail so that the appearance of the whole object can be contained in the bounding box. Table 2 shows the defined numbers according to the object and action class and their statistics.



Finally, the collected datasets were divided into the training and the test sets. Similar to the dataset composition of the human action detection field (Soomro et al., 2012), the collected YouTube video dataset was grouped into subsets based on the object and action class. Since a single video contains different types of construction equipment action, the videos were divided into clips and assigned to different subsets according to the object and action class. Particularly, clips that contained the duplicate actions from the same video were excluded to ensure the independence of training and test sets. To conduct threefold cross-validation while maintaining the distribution in the training and test sets, six clips were assigned to each subset and randomly divided into three groups of two clips each. Two of them were used for training and the rest one was used for testing of the model so that they would not duplicate in the training and test sets. Then, the fixed number of sequential frames (e.g., eight or 16 frames that had the same size of depth) in the same clip were combined to be the input for the training and testing models.

5.2 | Training and testing

Based on the collected dataset, training and testing were performed on ADVICE. A personal computer with two Intel Core i7-9700 @ 3.00 GHz, 32 GB DDR4, and one Nvidia RTX 2080Ti GPU was used for the training and testing. ADVICE is a one-stage and single-stream model, so the training and testing process were carried out on an end-to-end basis. Training and testing were conducted with four and two clips in each subset, and three-fold cross-validation was conducted to evaluate the proposed model.

First, ADVICE was trained on the 520 clips. The study applied transfer learning based on the weight parameters of the pre-trained 3D ResNeXt-101 model on the Kinetics dataset. The overfitting is prevented during training by using data augmentations. The setting for the model training is obtained as follows: (i) As the loss function, the mean squared error was used for the location elements of bounding boxes, and the binary cross-entropy was used for the confidence score and the probability of class; (ii) adaptive moment estimation was used as the loss optimization algorithm; (iii) the training rate started from 0.001 and gradually decreased; (iv) the batch size was set to 4; and (v) a total of 30 epochs of training were carried out.

Second, testing on the 260 clips was conducted. With ADVICE, the final output shows the bounding box regression results on all grid cells so that several bounding boxes on the same construction equipment can be predicted. Therefore, the study conducted the nonmaximum suppression combining the bounding boxes having the

same class among the duplicated bounding boxes with the intersection over union (IoU) at a certain level or higher. In this study, the possible actions differ by the type of construction equipment so the action class is subordinated to the object class. Subsequently, the nonmaximum suppression was performed on bounding boxes having the same object class. This way, accuracy can be improved by limiting the duplicated indication of the bounding boxes with the same construction equipment. The setting for the model testing is obtained as follows: (i) The bounding boxes whose object confidence score is below 0.75 were removed; (ii) the threshold of the IoU that performs the nonmaximum suppression was set to 0.5 or higher; and (iii) if the predicted class of the bounding box with the IoU of 0.5 or above is true, it is assumed to be true-positive.

As for the evaluation metric of the test, precision, recall, and F1 score were used. In particular, F1 score, considering precision and recall at the same time, is highly sensitive to imbalanced data, and it was used as an indicator for action detection evaluation in the field of computer science (Lea et al., 2017; Onal Ertugrul et al., 2019).

5.3 | Results and discussions

5.3.1 | Performance evaluation of ADVICE

ADVICE was developed for real-time action detection of various types of construction equipment on diverse videos recorded at different construction sites. Therefore, the detection scores and inference speed were analyzed as the main performance metrics. Particularly, the performance of the action detection model depends on the number of input frames, therefore the spatio-temporal features extracted from the single-stream feature extraction network are changed. The more the input frames, the more the start of the inference process is delayed, and the more the calculation and parameters increase, which make the inference speed slower. Therefore, this study compared the detection performance and inference speed of ADVICE by changing the number of input frames. Among the power of 2 that can be the input frames of the 3D ResNeXt-101, the two cases where the number of input frames is 25, corresponding to 1 s or below, are compared: 8 input frames with 1 interval (case 1) and 16 input frames with 1 interval (case 2; see Table 3 and Figure 6). Furthermore, it was not available to run the model in which the input was over 32 frames due to the memory limits of GPU.

First, the F1 scores of cases 1 and 2 in the object and action detection were compared while the IoU threshold was changing (see Figure 6). Regardless of the action class, when the predicted object class was correct, it is considered as a true-positive in the object detection task. On the

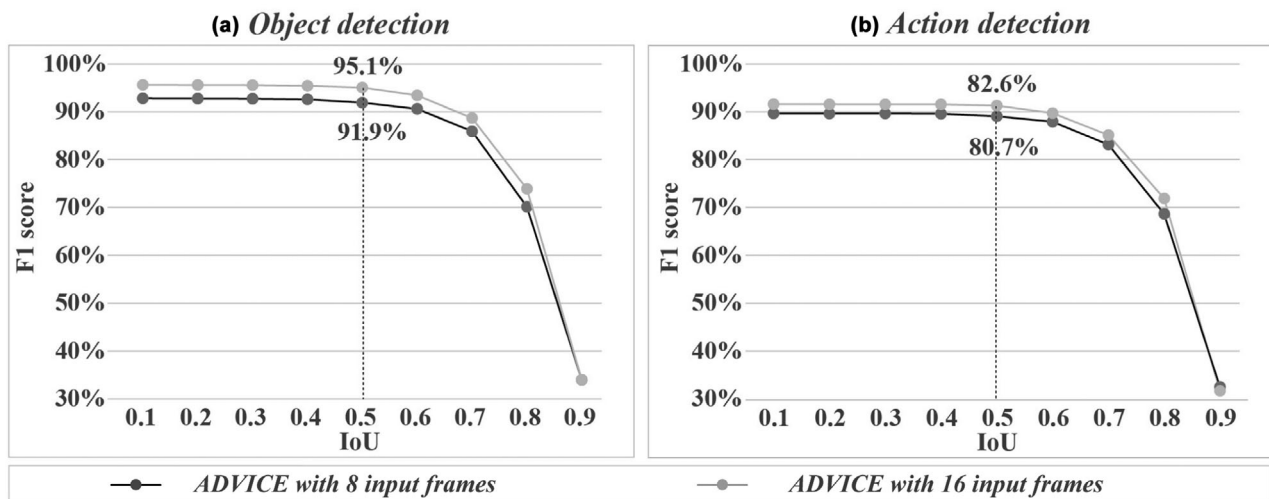


FIGURE 6 The performance of object detection (left) and action detection (right) according to the intersection over union

TABLE 3 Object and action detection performance according to the input frame numbers

Task	Case	Precision	Recall	F1 score
Object detection	1	91.1%	92.7%	91.9%
	2	94.5%	95.7%	95.1%
Action detection	1	80.0%	81.4%	80.7%
	2	82.1%	83.1%	82.6%

other hand, when both the object class and the action class were correct, it is considered as a true-positive in the action detection task. In both object and action detection tasks, the detection performance of case 2 was better than that of case 1 in all the IoU thresholds. Particularly, the F1 score of case 2 in the object and action detection was 2.7% higher and 1.5% higher on average in all the IoUs. Furthermore, when the IoU was at the common threshold, which is typically 0.5, the F1 scores of case 2 in object and action detection were 95.1% and 82.6%, which are both higher than 91.9% and 80.7% of case 1.

Second, performance scores in the object and action detection were compared (see Table 3). In the object detection, case 2 showed higher precision and recall, as well as a higher F1 score of 95.1% than case 1. In the action detection, case 2 had a higher precision, recall, and F1 score than case 1 by 2.1%, 1.7%, and 1.9%, respectively, in which the performance differences are greater than in the object detection task.

Overall, the case with 16 input frames showed improved performance in both object and action detection than the case with eight input frames. However, the inference time of case 2 at 23.5 ms was longer than that of case 1 (see Table 4). Thus, if the action detection model is performed on a lower-end PC or with videos at 30 fps or more, case 1 can be used for real-time inference, despite the perfor-

TABLE 4 Inference time according to the input frame numbers

Case	Inference time (ms)	Frames per second
1	23.4	42.7
2	27.3	36.6

mance trade-off with the shorter input frames. In this study, ADVICE with 16 input frames was selected as the optimal model to compare with other models.

5.3.2 | Ablation study

The study applied the 3D attention module and the feature pyramid network to the developed ADVICE. Furthermore, the multiclass output layers were proposed to separately predict the probability of object type and its action. To determine their effectiveness for real-time action detection of the construction equipment, the ablation study was conducted. Therefore, ADVICE was compared with two different models: (i) 3D ResNeXt-101 without the two algorithms using the multiclass output layer (i.e., the basic model); and (ii) 3D ResNeXt-101 with the two algorithms using the binary-class output layer (i.e., the binary-class model) in which the total of 13 actions of four construction equipment were considered as a binary class in order. For instance, the construction equipment labeled “2” (i.e., truck) for object class and “2” (i.e., travel) for action class in the multiclass output is transformed into “7” (i.e., truck-travel) in the binary-class output.

First, the object and action detection performance of ADVICE and the basic model were compared (see Figure 7). Overall, ADVICE showed better performance than the basic model on both object and action detection. The precision, recall, and F1 score in all the test

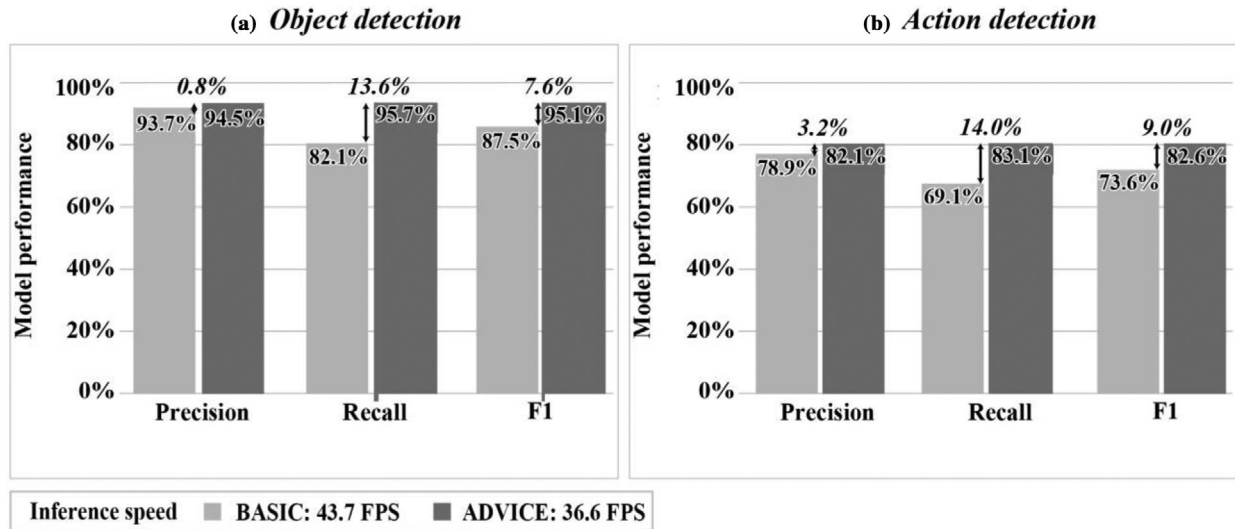


FIGURE 7 The performance of the basic model and ADVICE on object detection (left) and action detection (right)

TABLE 5 F1 scores according to the object size of the basic model and action detection in video of construction equipment (ADVICE)

Object size (bounding box area)	Small (under 12,000)	Medium (12,000 to 39,420)	Large (39,420 or above)
Basic model	64.5%	80.5%	75.6%
ADVICE	82.1%	84.5%	81.1%

datasets were 0.8%, 13.6%, and 7.6% higher on object detection, respectively. Furthermore, the precision, recall, and F1 score on all the test datasets were higher by 3.2%, 14%, and 9% on action detection, respectively. Particularly, object detection performance improved considerably because of the improvement in small object detection with the implementation of the feature pyramid network. As shown in Table 5, the comparison of the action detection performance on three different dataset clusters, which divided the total test dataset into an equal number according to the size of the bounding box, showed that the F1 score on small objects improved by 17.6%, which showed a remarkable performance improvement over the medium and big object detection. Meanwhile, the difference in performance increased more in the action detection task than in the object detection task. This shows that the 3D attention modules applied to the 3D-CNN architecture were also effective in extracting the spatio-temporal features for action detection. The performance reduced as the object size increases are possible since the number of clips of a specific action label may belong to a specific group.

Second, the object and action detection performance of the ADVICE and the binary-class model were compared (see Table 6 and Figure 8). Overall, ADVICE showed bet-

TABLE 6 F1 scores of the binary-class model and ADVICE

Object detection		Action detection	
Binary-class model	Advice	Binary-class model	Advice
85.7%	95.1%	72.9%	82.6%

ter performance than the binary-class model in both object and action detection. As shown in Table 6, the F1 score in all the test datasets was 9.4% higher in object detection. Meanwhile, the F1 score of ADVICE in action detection for all the datasets 9.7% higher. Furthermore, as shown in Figure 8 which represents the confusion matrix of the number of targets and predicted labels with F1 scores, the object and action detection performance of ADVICE improved, compared to the binary-class model in all object and action classes. As a result, it is validated that the multiclass output layer could improve the detection performance by separately estimating probabilities of object and action class, which could affect each other when they were considered as a binary class. Thus, ADVICE showed better performance not only in the classification of the object in the same action but also in the classification of the action in the same object, which is one of the challenging factors of action detection in construction sites. It is possible when the number of clips of a specific action and object label may belong to a specific size group, compared to other size groups.

Overall, the 3D attention module and the feature pyramid network were validated to improve the object detection performance of the developed model, ADVICE. In addition, the application of the multiclass output layers to the 3D CNN-based one-stage model could enhance the detection of the object and action of construction

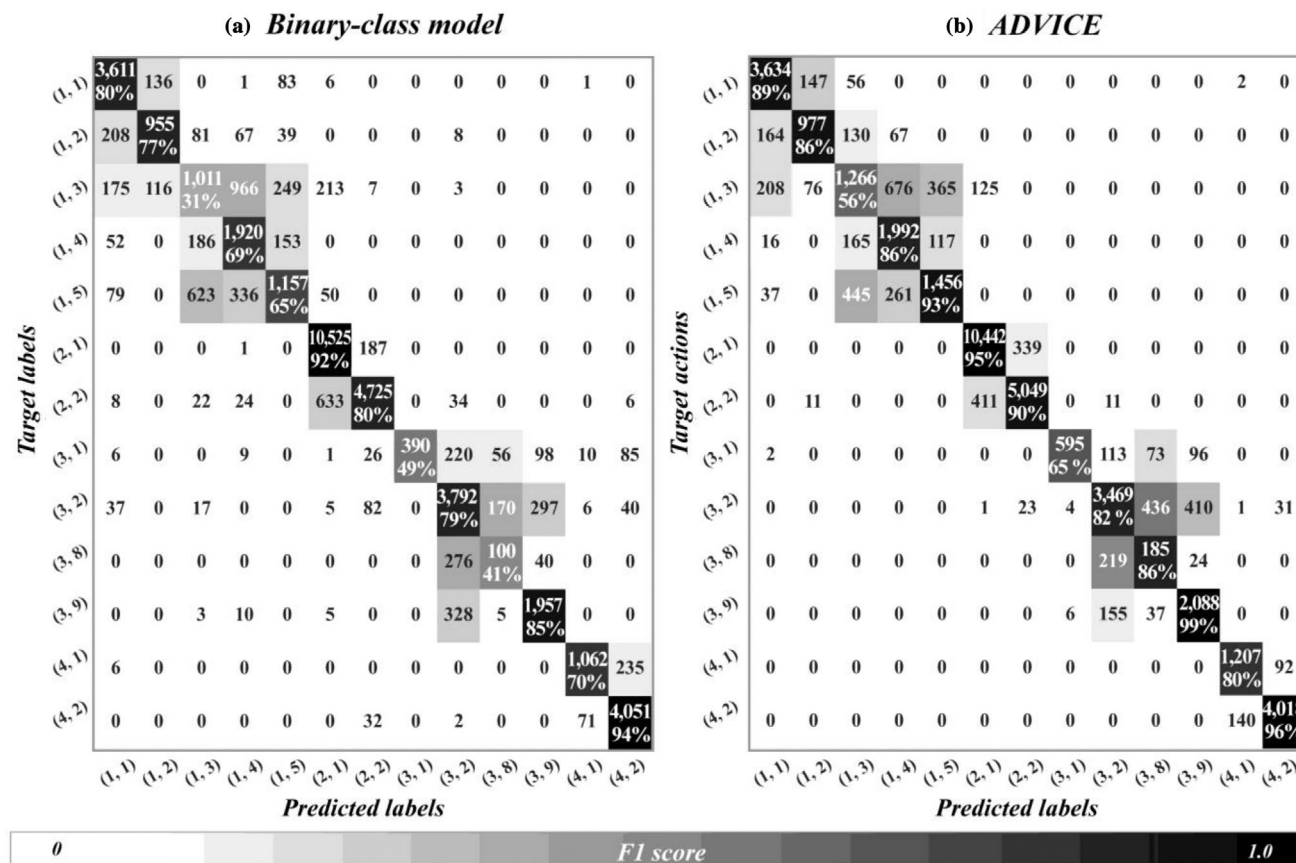


FIGURE 8 Action detection performance of the (a) binary-class model and (b) ADVICE. The numbers and the percentage scores in each grid represent the number of the corresponding target and predicted labels and F1 scores on each class

equipment. Although the application of the attention modules and the feature pyramid network reduced the inference speed of ADVICE, which increased the inference time, 25 or more frames could still be processed within 1 s (see Figure 7).

5.3.3 | Comparative analysis of ADVICE with the multistage model

Unlike the multistage models proposed in several previous studies (Chen et al., 2020; J. Kim & Chi, 2019; Luo et al., 2020; Roberts & Golparvar-Fard, 2019), ADVICE is a one-stage model in which the object and action detection tasks are performed at once. Therefore, to evaluate the effectiveness of ADVICE, it was compared with the existing multistage model. As in the previous studies, the multistage model consists of three stages: (i) Object detection using YOLOv3, (ii) tracking using DeepSort based on the bounding box of the object detection, and (iii) action classification using 3D ResNext-101 by entering the sequential images cropped by the minimum bounding box that can encompass all tracked bounding boxes of the same object

TABLE 7 Performance of the algorithms in the multistage model trained with the same datasets

Task	Model	Precision	Recall	F1 score
Object detection	Yolov3 + DeepSort	94.8%	91.1%	92.9%
Action classification	3D ResNext-101	90.8%	90.7%	90.7%

in sequential frames. Particularly, the 3D CNN algorithm was adopted to the multistage model since it outperformed the other methods such as 2D CNN only or 2D CNN with LSTM in the action classification task (Hara et al., 2018; Ji et al., 2013; Liu et al., 2018; Lu et al., 2018). All the algorithms are trained on the same datasets used for training ADVICE, and their performance is shown in Table 7.

First, the object detection and action detection performance were analyzed (see Table 8). In the object detection, the recall and F1 score of ADVICE were 4.6% and 2.2% higher than those of the multistage model, respectively, despite the precision that was 0.3% lower. Moreover, in the action detection, the precision, recall,



TABLE 8 Detection performance of multistage model and ADVICE for all dataset

Task	Model	Precision	Recall	F1 score
Object detection	Multistage	94.8%	91.1%	92.9%
	ADVICE	94.5%	95.7%	95.1%
Action detection	Multistage	56.5%	54.3%	55.4%
	ADVICE	82.1%	83.1%	82.6%

and F1 scores were 25.6%, 28.8%, and 27.2% higher than those in the multistage model, respectively. In particular, although each algorithm used in the multistage model had over 90% of performance scores, respectively, the precision, recall, and F1 scores declined to 56.5%, 54.3%, and 55.4%, when the sequential images predicted in the object detection and tracking process were given to the action classification model. This performance decline was due to the separate process of the multistage model in which input of the action classification model was generated from the result of the object detection and tracking. The action classification accuracy of the multistage model highly depended on the accuracy of the bounding boxes generated in the tracking process. Particularly, there were input images containing only partial scenes of the action even if the object detection result was true-positive by exceeding the IoU of 0.5 after the tracking. As a result, the poor action classification performance for those inputs degraded the overall action detection performance.

Second, to evaluate the performance of the challenging factors (i.e., the occlusion, lighting, and camera motion), a comparative analysis was also conducted (see Figure 9). To this end, clips with occlusion, clips with weak saturation by lighting, and clips shot by a camera with motion were analyzed separately. In the results of the object detection, the precision was 1.4% and 3.4% higher in the multistage

model in terms of the occlusion and the camera motion, respectively. However, in the F1 score that reflected the recall performance, the performance of ADVICE was as high as 4.0%, 2.7%, and 0.4% in occlusion, lighting, and camera motion, respectively. As for the results of the action detection, ADVICE showed better performance in occlusion, lighting, and camera motion. The performance differences in the occlusion and lighting conditions were larger than the average performance difference that was 27.2%. In contrast, the performance difference in the camera motion (i.e., 23.3%) was smaller than the average performance difference. This is because the sequential frames are cropped focusing on the object by the bounding box from the object detection and tracking process, so moving of surrounding materials and the background by camera motions are not included as spatio-temporal information.

Overall, ADVICE showed better performance than the multistage model. Although the precision, recall, and F1 score were more than 90% in each stage of the multistage model, it showed lower performance than ADVICE when both the action and the object classes were considered. Since each step proceeded separately in the multistage model, the errors in each step could be accumulated, and the final performance was shown to be poor. Besides, ADVICE, as a single-stage model, predicts the entire context in the entire frame, whereas in the multistage model, the action classification model could not consider the context because it considers only the information in the bounding box. Moreover, in terms of the inference time, the multistage model showed a slower speed (i.e., 53.8 ms and 18.5 fps) than the one-stage model because the multistage model proceeds through each step in order and extracts the results. As a result, the ADVICE model performed with higher accuracy in real time than the multistage model.

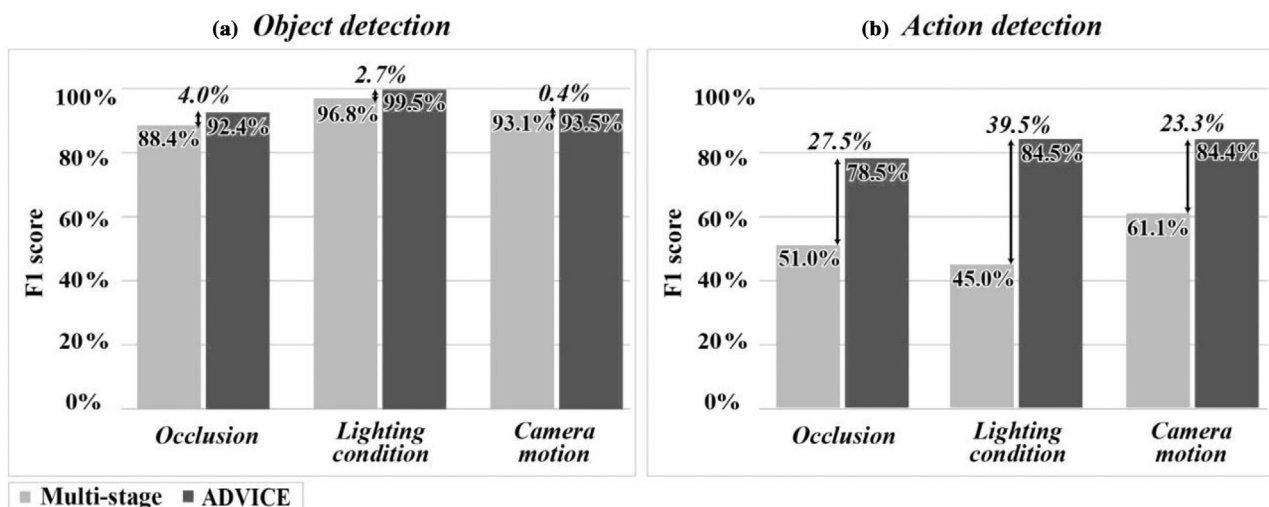


FIGURE 9 Object and action detection performance of challenging factors (i.e., occlusion, lighting condition, and camera motion)



5.3.4 | Discussions

The diversity, variability, and complexity of the construction site are important and challenging factors that need to be resolved when implementing a computer vision-based method in the construction industry. When compared to previous studies that proposed a computer-vision based method for action detection of construction equipment (Chen et al., 2020; J. Kim & Chi, 2019; Roberts & Golparvar-Fard, 2019), the results obtained from the experiment validated the real-time applicability of ADVICE to a construction site with diversity, variability and complexity.

First, in contrast to the methods proposed by the previous studies, ADVICE can perform the detection process from the data input via feature extraction through a single-stage structure. To this end, the 3D attention module, feature pyramid network, and multiclass output layers are developed to improve the detection performance with negligible overhead computation. This computational novelty allows the developed model to detect the action detection in real time. Furthermore, end-to-end training and testing are possible as well.

Second, in contrast to the previous studies that performed action detection only on a maximum of two types of construction equipment (excavator and truck), this study performed action detection by targeting the seven types of actions from the four types of construction equipment. This allows for the action detection of the diverse types of construction equipment.

Third, in contrast to the previous studies that used less diverse types of videos of construction sites, this study collected a total of 130 videos recorded in different conditions, which were then used for model training and testing. Thus, it allows for robust action detection under variable site conditions. In addition, the proposed model showed higher action detection performance even in challenging situations such as occlusions, different lighting conditions, and recording with camera motions, which were included in the datasets.

Last, the action classification performance of the computer-vision-based construction equipment action detection model proposed in the previous studies was between 80% and 90% of recall rates. Such performance results may decrease further if the accuracy of the object detection and tracking, the stage previous to the action classification stage, is included such as the multistage model that showed recall rates between 50% and 60% in this study. On the other hand, the precision and recall on all test sets of ADVICE were 82.1% and 83.1%, respectively. Considering the higher number of classes and the possibility of real-time action detection, the error rates can be acceptable, which shows the computational novelty.

In terms of the detection result on a class “loader, dig,” compared to other classes, the precision, recall, and F1 score showed low scores at 35.9%, 42.1%, and 38.8%, respectively, which indicates the limitation of this study (see Figure 8). The number of bounding boxes on the four types of construction equipment was around 130,000, but the digging loader is about 1675, which was about 1% of the total bounding boxes. Thus, compared to the other action classes, the training on the diverse video frames recorded in different conditions and construction sites was lacking, which may be the reason for the lower accuracy. Similar to the other classes, the evaluation performance of the model may be further improved with sufficient datasets. Another limitation is that there are construction equipment actions that cannot be distinguished by visual information. For instance, in the case of working actions underneath the surface, such as an auger or breaker, accurate classification of the action class of those types of construction equipment via video frames is limited. In such cases, action detection can be performed by fusing visual information with other types of data, such as audio information.

6 | CONCLUSION

A 3D CNN-based one-stage model for real-time action detection in videos of construction equipment is developed focusing on real-time monitoring of various types of construction equipment: excavator, truck, loader, and roller. To overcome the limitations of previous studies that monitor construction equipment using multistage model, ADVICE is designed with the implementation of the 3D attention module and the feature pyramid network developed in this study to refine the spatio-temporal features. Moreover, the output layer was designed for multiclass detection problems (i.e., object class and action class). A total of 130 videos were collected to validate various actions of construction equipment concerning occlusion, lighting, scale of objects, and motions of the camera.

Based on the training in the experimental study, ADVICE achieved 82.1% precision and 83.1% recall with an inference speed of 36.6 fps on all test sets. The comparative analysis showed that the average F1 scores in ADVICE were 9% and 9.7% higher than in the basic model without the 3D attention module and the feature pyramid network. In the binary-class model, particularly, compared to the existing multistage model, ADVICE showed 27.2% higher F1 score maintaining decent performance even in challenging situations such as occlusion, different lighting conditions, and camera motions included.

This study is the first study dealing with the action detection of construction equipment using the 3D CNN-based one-stage model. Particularly, by implementing the



3D attention module and the feature pyramid network, ADVICE can detect the complex actions of multiple construction equipment with different scales in real time.

A distinct advantage of ADVICE is that it can detect construction equipment action based on the surveillance video, instead of utilizing human resources or expensive monitoring sensors. Furthermore, ADVICE can be applied to videos with various camera motions from a hand-held camera or an unmanned aerial vehicle, which is being increasingly utilized for construction management recently. It is anticipated to increase accuracy and speed with the application of the advanced neural network algorithm that is effective for the classification and recognition process (Ahmadlou & Adeli, 2010; Alam et al., 2020; Pereira et al., 2020; Rafiei & Adeli, 2017). Ultimately, ADVICE can be further generalized at construction sites by training with various types of equipment that were not considered in this study.

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